

This advanced HMD (Head Mounted Display) allowed the user to see all the graphics displayed by computers. HMDs could only be linked to cameras earlier and in 1966 they were connected to a computer for the first time.

What's surprising? Even at that point of time he built something that could be manipulated by the user. It seemed that the environment was in control of the user and not the computer. The HMD tracked the user's head movements changing his focal view accordingly.

Courtesy to Sutherland's Ultimate Display, a brand new virtual world is on its rise. Now that we have already stepped into this simulated world, let's dig deeper.

How do you think any structure stands tall? You don't have to scratch your brains to put some techno-logic to it. With its pillars, of course! If you are already mind blown by VR, here's how it works. Below are few concepts on which VR still continues to stand tall.

## 1. Simulation

How do you think it will be possible to land on Mars while you are about to bump onto your bedroom wardrobe? It all depends on simulation! Simulation is creating a virtual environment that closely resembles the real world. To simulate something, a model has to be created mimicking the key characteristics of the real world. This can be as simple as voice recognition or tracking the body movements.

In VR, simulation is created using a computer. The computer-simulated environment will be manipulated by the user with HMDs mounted on the face.

The immediate question that arises after creating such virtual environments is that how a person will interact with it?

## 2. Interaction

Interaction- another concept of VR unfolds the answers right here. Navigating inside a simulated environment itself is called interaction. Interacting with the artificial environment is possible only when the user is completely immersed in it.

Interactions need manipulations and here, it is the user that holds the key to manipulate. Interactivity depends on factors like range, speed and mapping. Mapping is the ability of the system to produce results based on user's actions.